Mesh15 Protocol Specification

A MANET Digital Voice and Data Protocol for Handheld Radio

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*This document specifies the Mesh15 protocol. The keywords **SHALL**, **SHALL NOT**, **MUST**, **REQUIRED**, **RECOMMENDED**, **SHOULD**, and **MAY** are to be interpreted as described in RFC 2119.*

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1. Introduction

1.1 Purpose

This document specifies the **Mesh15** protocol, a fully distributed, multi-hop, real-time digital voice and data networking protocol intended for handheld radio hardware built around the Texas Instruments CC1200 sub-gigahertz transceiver. The specification is written to a level of detail sufficient for independent implementation by multiple parties on different software and hardware platforms that share the CC1200 family of devices.

1.2 Scope

The specification covers the following protocol layers and operational aspects:

- Physical layer configuration for the CC1200 transceiver family.
- Forward error correction, scrambling, and interleaving.
- Cryptographic layer including payload AEAD, key hierarchy, and identity.
- Frame layout, addressing, identifier allocation, and lifetime management.
- Medium access control.
- Routing and route maintenance.
- Voice service, including Codec2 voice transport and group semantics.
- Data service with reliable datagram semantics.
- Group completion protocol, including confirmation and repair.
- Operations, administration, and maintenance procedures.
- Naming, versioning, and compliance profiles.

1.3 Relationship to Other Standards

Mesh15 is an independent protocol. It is not a derivative of M17, DMR, or any other digital voice standard. The payload rates of Codec2 in Mesh15 are aligned with the canonical Codec2 3200 and 1600 modes used in M17, but the air interface, the link layer, and the routing layer are entirely Mesh15 specific. A future annex will define an optional M17 interoperation profile.

1.4 Conformance

Conformance to this specification is defined in Section 18. Implementations claiming Mesh15 compliance at a given level SHALL satisfy all of the mandatory normative requirements associated with that level.

1.5 Document Conventions

The keywords **SHALL**, **SHALL NOT**, **MUST**, **MUST NOT**, **REQUIRED**, **RECOMMENDED**, **SHOULD**, **SHOULD NOT**, and **MAY** are to be interpreted as described in RFC 2119.

All multi-byte numeric fields are encoded in **network byte order** (big-endian) unless otherwise noted.

Signed integers are two's complement.

2. Definitions, Acronyms, and Conventions

2.1 Definitions

Term	Definition
Call	A logical conversation that may be unicast or multicast and may consist of o
Codec2	The open-source low bit-rate speech codec by David Rowe.
Epoch	A single block of voice data derived from a fixed-duration audio frame.
Group	A logical membership set identified by a Group Identifier.
Group Completion	The protocol by which the talker confirms, not-guarantees, that members of
HAVE Vectors	Compact bitmaps of received epochs exchanged between nodes.
Hop	A single Layer-3 forwarding of a frame from one Mesh15 node to its success
Hop ACK	A Layer-2 acknowledgement confined to the link between two adjacent node
Mesh	A general topology in which any node may forward to any other node.
Node	A single Mesh15-capable device.
Path	A sequence of nodes from source to destination.
Repair	A retransmission of one or more lost epochs subsequent to their first transm
Spur	A continuous talkspurt initiated by PTT press and terminated by PTT release
Talker	The node currently transmitting voice.
Transceiver	The CC1200 or compatible device.

2.2 Acronyms

Acronym	Expansion
AEAD	Authenticated Encryption with Associated Data
AGC	Automatic Gain Control
ARQ	Automatic Repeat Request
CCA	Clear Channel Assessment
CRC	Cyclic Redundancy Check
CSMA	Carrier Sense Multiple Access
DA	Destination Address
DV	Digital Voice
ETX	Expected Transmission Count
FEC	Forward Error Correction
FIFO	First In First Out
GCD	Group Completion Delay
GCS	Group Completion Sequence
HT-15	The Arkos Engineering HT-15 handheld radio platform

Acronym	Expansion
LNA	Low Noise Amplifier
MAC	Medium Access Control
MCS	Modulation and Coding Scheme
MIC	Message Integrity Code
OGM	Originator Message
PA	Power Amplifier
PCM	Pulse Code Modulation
PDU	Protocol Data Unit
PER	Packet Error Rate
PHY	Physical Layer
PLC	Packet Loss Concealment
PTT	Push-to-Talk
PSK	Pre-Shared Key
RLC	Random Linear Code
RSSI	Received Signal Strength Indicator
RX	Receive
SA	Source Address
SNR	Signal-to-Noise Ratio
SPI	Serial Peripheral Interface
TDMA	Time Division Multiple Access
TX	Transmit
VOPEN	Voice Open, the call-set signaling frame

2.3 Notation

The following notation is used throughout this document:

- Bit positions numbered from 0 (most significant) within a byte unless otherwise stated.
- Frame fields are documented in the order they appear on the air.
- Tables of register values default to hexadecimal unless prefixed with `0b` or otherwise indicated.
- Arithmetic with named signals follows the convention documented in the relevant section.

2.4 Reference Time

All durations are expressed in microseconds (μs), milliseconds (ms), or seconds (s). The reference epoch for transmit and receive timing is the local clock of the node, synchronized to its neighbors by the beacon and probe procedures of Section 10.

3. System Overview

3.1 Goals

Mesh15 is designed to satisfy the following primary objectives:

1. **Voice-first operating model.** Voice SHALL be the principal service of the protocol. Data is supported as a peer service.
2. **Deterministic multi-hop real-time voice.** A voice spur originating at a node SHALL be audible at any group member for which a route exists, with bounded end-to-end latency.
3. **Distributed topology.** No node SHALL be required for any function other than as a member of the mesh. There is no coordinator, base, or repeater.
4. **Hardened delivery semantics.** The talker SHALL be informed of delivery confirmation status for the group. Failures SHALL be reported rather than silently omitted.
5. **Cryptographic security.** Payload confidentiality and integrity SHALL be available as a first-class feature.
6. **Peer auto-discovery.** A node that joins the network SHALL become routable without pre-programmed knowledge of other nodes.
7. **Scalability.** The protocol SHALL scale to at least 32 nodes per group with full voice QoS and SHALL continue to operate with degraded service at higher counts.

3.2 Non-Goals

The following are explicitly out of scope for Mesh15:

- Connection-oriented streaming with TCP-style retransmission for voice.
- Guaranteed delivery under hidden terminal conditions that prevent the talker from ever being heard.
- Inter-operation with proprietary protocols that have not been publicly documented.
- Implementations that rely on a central coordinator, base station, or single point of failure.

3.3 Service Model

A node provides three services:

- **Voice Service.** Push-to-talk group or unicast voice using Codec2.
- **Data Service.** Reliable datagram transport with optional streaming semantics.
- **Completion Service.** Distributed tracking and repair of voice epochs to guarantee group confirmation or explicit failure.

These services are independent and concurrent. The scheduler arbitrates between them as described in Section 9.

3.4 Topology

Mesh15 supports arbitrary mesh topologies. The following topology classes are recognized:

- **Single hop.** A talker and group members within radio range.
- **Linear chain.** A and B within range, B and C within range, A and C not within range.
- **Branching tree.** A talker with multiple groups forwarded along different next hops.
- **Graph.** Multiple redundant paths. The protocol uses primary-next-hop with backup-next-hop resolution.

All nodes are peers. Any node may originate a call. Any node may forward frames for any other node.

3.5 Reference Network Sizes

Class	Maximum Guaranteed Voice Members	Engineering Notes
Small	16	Default engineering target.
Medium	32	Full-featured voice QoS.
Large	64	Voice QoS is best-effort above 32.
Unbounded	No hard limit	Service degrades gracefully.

4. Reference Architecture

4.1 Layer Model

The Mesh15 architecture is composed of the following layers. Each layer SHALL be implemented as a logical component and SHALL communicate with adjacent layers only through well-defined interfaces.

Layer	Function
Application	PTT, user interface, key management, identity provisioning.
Completion	Tracking of group HAVE state, computation of confirmation, unkey repair.
Voice	Codec2 encoding, voice spur state, jitter buffer, repair eligibility.
Data	Reliable datagram window, fragmentation, reassembly.
Net	Routing tables, path selection, multipath maintenance, OGM processing.
Link	Frame construction, hop ARQ, AEAD verification, MCS selection.
MAC	Slot arbitration, VOPEN floor, control and data window scheduling.
PHY	CC1200 configuration, transmit and receive state machine, RSSI, CCA.

4.2 Core Data Path

The transmit path proceeds as follows:

1. Application layer receives audio samples from the codec.
2. Voice layer encodes the audio into a Codec2 frame.
3. Completion layer creates or extends a spur record and assigns an epoch sequence number.
4. Link layer packages the audio data with FEC, header, and AEAD.
5. MAC layer requests a transmit opportunity.
6. PHY layer programs the CC1200 and transmits the frame.

The receive path proceeds in reverse order. At the MAC layer, the receive scheduler dispatches control frames to the Link layer immediately and forwards voice and data frames to the appropriate higher layer queue.

4.3 Reference Node Block Diagram

The reference implementation that motivates this specification is the Arkos Engineering HT-15 platform. The block diagram is as follows:

```
+----------------+      +-------------------+
| TLV320 Audio   |      | SSD1681 Display   |
| Codec (DAC3100)|      | (E-Ink, 250x122)  |
+-------+--------+      +---------+---------+
        | I2S                     | SPI
        v                         v
+----------------------------------------------------+
|                       RP2350                       |
|  +-----------+  +-----------+  +------------+     |
|  | AUDIO     |  | RADIO     |  | APP / UI   |     |
|  | CORE 0    |  | CORE 1    |  | CORE 1     |     |
|  +-----+-----+  +-----+-----+  +------------+     |
|        |              |                            |
|        |              | SPI / DMA                  |
|        |              v                            |
|        |       +--------------+                    |
|        |       |  CC1200 RF   |                    |
|        |       |  Transceiver |                    |
|        |       +------+-------+                    |
+-------|--------------|---------------------------+
        |              | RF
        |              v
+--------+      +-----------------+
|  Mic   |      | External PA     |
| Path   |      | 3.5 W Output    |
+--------+      +-----------------+
```

The protocol is independent of the platform insofar as the M-class (Mesh15 compliance) implementation requires only that the platform provide a CC1200 or compatible transceiver under the control of a compute subsystem capable of running the link and net layers.

4.4 Identifier Conventions

Three distinct identifier types are used in Mesh15:

- **Mesh15 Node Identifier (NID).** A 32-bit identifier unique to the device. NID is derived from the device identity key.
- **Mesh15 Short Alias (M16).** A 16-bit local alias assigned dynamically through the discovery protocol.
- **Group Identifier (GID).** A 16-bit identifier naming a logical multicast group.

These identifiers are defined more precisely in Section 8.

5. Physical Layer

5.1 General

The Mesh15 PHY is implemented on the CC1200 transceiver family. Configurations are normative where they are required to interoperate; alternative configurations are permitted only when explicitly identified as such.

The CC1200 is a half-duplex transceiver. A node SHALL NOT assume that it can transmit and receive simultaneously. All procedures in this specification are written under the half-duplex assumption.

5.2 Frequency Bands

The Mesh15 PHY supports the following frequency bands as defined by the CC1200 synthesizer:

Band	Range	Typical Use
2 m	136 to 174 MHz	Default range-optimized band.
70 cm	410 to 480 MHz	Default bandwidth-optimized band.
33 cm	820 to 960 MHz	Permitted where regulated.

Field deployments SHALL respect the applicable regulatory domain. The protocol does not impose regulatory constraints beyond interoperability.

5.3 Modulation and Coding Scheme Ladder

Mesh15 defines a numbered set of MCS values. Each MCS has a name, a modulation format, a symbol rate, an occupied bandwidth, and an inclusive maximum net throughput.

MCS	Name	Modulation	Symbol Rate (sym/s)	Bit Rate (b/s)	Occupied BW (kHz)	Coding
0	MCS-R0	2-GFSK	1,200	1,200	12.5	Rate 1/2 FEC
1	MCS-R1	2-GFSK	2,400	2,400	12.5	Rate 1/2 FEC
2	MCS-R2	2-GFSK	4,800	4,800	12.5	Rate 1/2 FEC
3	MCS-R3	2-GFSK	9,600	9,600	15.0	Rate 1/2 FEC
4	MCS-R4	4-GFSK	4,800	9,600	12.5	Rate 1/2 FEC
5	MCS-M0	2-GFSK	19,200	19,200	25.0	Rate 1/2 FEC

MCS	Name	Modulation	Symbol Rate (sym/s)	Bit Rate (b/s)	Occupied BW (kHz)	Coding
6	MCS-M1	2-GFSK	38,400	38,400	50.0	Rate 1/2 FEC
7	MCS-M2	2-GFSK	50,000	50,000	100.0	Rate 1/2 FEC
8	MCS-B0	2-GFSK	100,000	100,000	200.0	Rate 2/3 FEC
9	MCS-B1	2-GFSK	250,000	250,000	400.0	Rate 2/3 FEC
10	MCS-F0	2-GFSK	500,000	500,000	800.0	Rate 3/4 FEC
11	MCS-F1	2-GFSK	1,000,000	1,000,000	1,500.0	Rate 3/4 FEC

Norms:

- MCS values 0 through 4 are collectively designated the **range profile** family.
- MCS values 5 through 7 are designated the **mesh profile** family.
- MCS values 8 and 9 are designated the **burst profile** family.
- MCS values 10 and 11 are designated the **fast profile** family.

Implementations SHALL support at least MCS-R0, MCS-R3, MCS-M0, and MCS-M1.

5.4 MCS Selection

The selection of an MCS for a given link is performed by the link layer. The procedure is as follows:

1. Each node SHALL maintain a per-neighbor metric consisting of running estimates of:

a. Packet Error Rate (PER) over the last 64 PDUs.

b. Average RSSI.

c. Standard deviation of RSSI.

d. Age of last successful receive.

2. The link layer SHALL periodically probe a higher MCS on each active neighbor link. The probe SHALL be a small data PDU of size not exceeding 16 bytes.
3. If the probe succeeds over a window of 8 PDUs, the link layer MAY upgrade to the next-faster MCS.
4. If the running PER exceeds 25%, the link layer SHALL downgrade to the previous-slower MCS.
5. The MCS used for a given transmit is encoded in the per-link frame header.

5.5 CC1200 Configuration

This section defines the normative configuration of the CC1200 for each MCS. Implementations SHALL program the corresponding registers at session establishment and at any subsequent MCS change.

The configuration uses the following build blocks:

- **Frequency offset.** Set via `FREQ2`, `FREQ1`, `FREQ0` for the carrier.
- **Symbol rate.** Set via `SYMBOL_RATE2`, `SYMBOL_RATE1`, `SYMBOL_RATE0`.

- **Modulation format.** Set via `MODCFG_DEV_E`.
- **Deviation.** Set via `DEVIATION_M` and `MODCFG_DEV_E`.
- **Receive channel bandwidth.** Set via `CHAN_BW`.
- **Modem behavior.** Set via `MDMCFG1`, `MDMCFG0`.
- **AGC.** Set via `AGC_REF`, `AGC_CS_THR`, `AGC_CFG1`, `AGC_CFG0`.
- **Preamble and sync.** Set via `PREAMBLE_CFG1`, `PREAMBLE_CFG0`, `SYNC_CFG1`, `SYNC_CFG0`, `SYNC3`, `SYNC2`, `SYNC1`, `SYNC0`.
- **Packet framing.** Set via `PKT_CFG2`, `PKT_CFG1`, `PKT_CFG0`, `PKT_LEN`.
- **PA configuration.** Set via `PA_CFG1`, `PA_CFG0`.
- **Frequency synthesizer.** Set via `FS_CFG`.

The detailed register values for each MCS are provided in Appendix A.

5.6 Sync Word

Mesh15 uses a 32-bit sync word as the frame preamble marker. The sync word is:

```
0x55 0xF7 0xA5 0x3C
```

The sync word SHALL be transmitted most-significant byte first. The sync word is followed by the length field and the frame body.

5.7 Preamble

The preamble length is adaptive. The default nominal preamble is 16 nibbles (8 bytes) of `0x55`. Implementations MAY extend the preamble to 24 or 32 nibbles when the receiver reports poor acquisition under the MCS link-state estimator.

5.8 Clear Channel Assessment

The CC1200 exposes a programmable CCA indicator via the `AGC_CS_THR` register and the `IOCFGx` GDO configuration. The MAC layer SHALL use CCA to defer transmissions when the channel is reported busy.

CCA SHALL be performed with a measurement window of 8 symbol periods. If the channel is reported busy, the MAC retries after a CSMA backoff as described in Section 9.

5.9 Receiver State

The CC1200 SHALL be configured for transparent mode operation with the CFM data path enabled for analog-like FM traffic. For packet voice, MCS-R3 and above, the packet frame mode is used with the FIFO threshold set to the half-depth of the FIFO (typically 64 bytes).

Two GDO pins SHALL be configured:

- One GDO pin SHALL be configured to assert on the rising edge of the sync word detect.

- One GDO pin SHALL be configured to assert on the packet done or FIFO threshold event.

The GDO pins are used to drive DMA requests on the host microcontroller.

5.10 Transmitter State

The transmitter SHALL be configured for maximum ramp-up smoothness in the 2 m and 70 cm bands. The `PA_CFG1` register SHALL be configured for the corresponding power amplifier. The default output power is +33 dBm conducted at the antenna connector.

5.11 Channel Selection

A node SHALL scan the operating channel upon startup and SHALL attempt to synchronize to the strongest neighbor. The channel selection SHALL be persisted across resets.

5.12 Link Budget Requirements

Designers SHALL verify that the link budget for the chosen MCS satisfies the deployment requirements. The recommended baseline budget assumes:

- Conducted TX power: +33 dBm.
- RX antenna effective at handheld: -4 dBi.
- Body loss: 4 dB.
- RF chain loss: 1.5 dB.
- Fade margin: 10 dB (suburban).
- Channel bandwidth: as specified per MCS.

Under these assumptions, the expected one-hop range for 90% reliability in a typical handset-to-handset scenario is approximately 2 km on 2 m with MCS-M0 and approximately 1.1 km with MCS-M1. The exact range is environment-dependent.

6. Forward Error Correction, Coding, and Scrambling

6.1 Overview

The Mesh15 link layer provides three distinct coding mechanisms:

- **Inner FEC.** A rate-1/2, rate-2/3, or rate-3/4 convolutional code applied to the entire payload.
- **Outer CRC.** A 32-bit CRC applied to the plaintext payload prior to FEC.
- **Scrambling.** A pseudo-random XOR applied to the transmitted bitstream to ensure adequate DC balance and avoid spurious clock recovery effects.

The exact codes are tabulated in the following subsections.

6.2 Inner FEC

The inner FEC is a rate-compatible punctured convolutional code with the following mother code:

- Constraint length K = 7.
- Generator polynomials: G1 = 0o133, G2 = 0o171.

The puncture tables are:

Rate	Polynomials	Puncture Pattern
1/2	G1, G2	none
2/3	G1, G2	one of four punctured symbols removed
3/4	G1, G2	two of four punctured symbols removed

The implementations SHALL use a soft-decision Viterbi decoder with traceback depth of 32 bits.

6.3 Outer CRC

The outer CRC is a CRC-32 with the following polynomial:

```
x^32 + x^26 + x^23 + x^22 + x^16 + x^12 + x^11 + x^10 + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1
```

The CRC is computed over the plaintext payload including all headers up to but not including the AEAD MIC.

6.4 Scrambling

A length-127 maximal-length sequence generator with polynomial x^7 + x^6 + 1 SHALL be initialized to 0x7F at the start of each frame. The output bit is XORed with the transmitted bitstream.

The receiver SHALL reinitialize the scrambler using the same procedure.

6.5 Interleaving

Blocks of 96 bits SHALL be interleaved using a 12-by-8 block interleaver. The interleaver is reset at the start of each frame.

6.6 Header Protection

The 16-byte link-layer header is itself protected by a 16-bit CRC independent of the inner FEC. The header CRC polynomial is:

```
x^16 + x^12 + x^5 + 1
```

A frame whose header fails the CRC SHALL be discarded without further processing.

7. Cryptographic Layer

7.1 Goals

Mesh15 provides:

- **Confidentiality.** Payload content SHALL be protected by authenticated encryption.
- **Integrity.** Headers and payloads SHALL be protected by an integrity check.
- **Replay protection.** Each frame SHALL carry a monotonically increasing nonce.
- **Identity.** A node SHALL be identifiable by a stable cryptographic identity.

7.2 Primitives

The Mesh15 cryptographic layer uses the following primitives:

Function	Primitive
Symmetric encryption	ChaCha20
Authentication	Poly1305
AEAD combination	ChaCha20-Poly1305
KDF	HKDF-SHA-256
Hash	SHA-256
Public key	Ed25519
Identity derivation	BLAKE2b-256

The primitives are normative. Alternative primitives SHALL NOT be used in the Mesh15 protocol unless explicitly documented in an annex.

7.3 Key Hierarchy

A node stores the following keys:

Key	Derivation	Storage
Device Identity Key (DIK)	Ed25519 key pair generated at factory	Tamper-resistant storage
Device Master Key (DMK)	Derived from DIK via HKDF-SHA-256	Persistent storage
Group Pre-Shared Key (GPSK)	Provisioned at deployment	Persistent storage
Group Session Key (GSK)	Derived from GPSK and per-call salt	Volatile, refreshed per call
Per-Link Key (PLK)	Derived from GSK and link identifier	Volatile

The Device Identity Key is bound to the device at provisioning time. The Device Master Key is derived from the DIK and is rotated only on authorized provisioning.

7.4 Group Sessions

A group session is established for the duration of a single call spur. The session key is derived as:

```
GSK = HKDF(GPSK, salt = call_id || epoch_0, info = "MESH15-GSK-v1", L = 32)
```

where `call_id` is a 64-bit identifier and `epoch_0` is the sequence number of the first epoch of the spur.

7.5 Per-Link Sessions

A per-link key is derived from the group session key and the link identifier as:

```
PLK = HKDF(GSK, salt = peer_nid, info = "MESH15-PLK-v1", L = 32)
```

The link identifier is the concatenation of the local and peer NIDs in ascending order.

7.6 Frame Confidentiality and Integrity

Each payload frame is encrypted and authenticated using ChaCha20-Poly1305 with the following parameters:

- **Key.** PLK.
- **Nonce.** 96-bit value derived from `seq_number` and `link_direction` as follows:

`nonce = seq_number (64 least-significant bits) || link_direction (1 bit) || 0x00 (31 bits).`

- **Plaintext.** The unencrypted payload.
- **Additional Data.** The 16-byte link-layer header.

The 128-bit Poly1305 tag is appended to the ciphertext. The receiver SHALL verify the tag prior to releasing the plaintext.

7.7 Discovery Privacy

The link layer SHALL encrypt the source and destination identifiers in the inner header. The outer header contains only the cryptographic alias used for routing.

7.8 Compliance

The cryptographic layer is OPTIONAL in Mesh15. Implementations that omit the cryptographic layer SHALL mark themselves as "Mesh15-Plain" and SHALL be subject to the regulatory constraints of the deployment domain.

8. Framing, Addressing, and Identity

8.1 Frame Categories

There are eight frame categories:

Code	Name	Usage
0x00	Beacon	Discovery and OGM.
0x01	Probe	Link quality probe.
0x02	Path	Routing and path maintenance.
0x03	VOPEN	Voice open, call setup.
0x04	VFRAME	Voice data epoch.
0x05	VACK	Voice acknowledgement.
0x06	VEND	Voice end, summary.
0x07	DATA	Reliable datagram.
0x08	CTRL	General control.
0x09-0xFF	Reserved	Reserved for future use.

8.2 Frame Format

All frames SHALL conform to the following general structure:

```
+--------------------+--------------------+
| Preamble (variable) | (PHY)              |
+--------------------+--------------------+
| Sync Word (4 bytes) | (PHY)              |
+--------------------+--------------------+
| Length (1 byte)    | (PHY)              |
+--------------------+--------------------+
| Outer Header (8 B) | (MAC)              |
+--------------------+--------------------+
| Inner Header (16 B) | (Link)             |
+--------------------+--------------------+
| Payload (0-251 B)   | (Link)             |
+--------------------+--------------------+
| AEAD MIC (16 B)    | (Cryptographic)    |
+--------------------+--------------------+
```

The maximum frame length is 255 bytes inclusive of headers and trailer. The maximum payload length is 251 bytes.

8.3 Outer Header

The 8-byte outer header is laid out as follows:

Offset	Field	Size	Description
0	version	1 byte	Mesh15 version. Set to 0x01 for this version.
1	frame_type	1 byte	Frame category.
2	flags	1 byte	Reserved. Set to 0x00.

Offset	Field	Size	Description
3	mcs	1 byte	Modulation and Coding Scheme.
4	src_alias	2 bytes	M16 short alias of source.
6	dst_alias	2 bytes	M16 short alias of destination, or 0xFFFF for b

The outer header is transmitted in the clear. The frame type, MCS, and short alias are used by the MAC layer and the routing layer to forward the frame without decryption.

8.4 Inner Header

The 16-byte inner header is laid out as follows:

Offset	Field	Size	Description
0	src_nid	4 bytes	Source NID.
4	dst_nid	4 bytes	Destination NID, or 0xFFFFFFFF for broadcas
8	group_id	2 bytes	Group Identifier.
10	seq_number	4 bytes	Sequence number.
14	hop_count	1 byte	Current hop count.
15	flags	1 byte	Frame flags.

The inner header is part of the AEAD additional data. It is encrypted only in the sense that it is authenticated.

8.5 Node Identifier

The Node Identifier (NID) is a 32-bit value derived from the device identity key as:

```
NID = lower_32_bits(BLAKE2b-256(DIK_public_key))
```

The NID is globally unique to the device.

8.6 Short Alias

The Short Alias (M16) is a 16-bit value assigned dynamically by the discovery protocol. The default initial value is the lower 16 bits of the NID. The alias is considered settled when no collision is observed for 30 seconds.

8.7 Group Identifier

The Group Identifier (GID) is a 16-bit identifier. GID values 0x0000 and 0xFFFF are reserved. The default groups are:

GID	Name	Notes
0x0001	All Call	Default group upon startup.

GID	Name	Notes
0x0002	Tactical A	Default tactical group.
0x0003	Tactical B	Optional.
0x0004-N	User-defined	Reserved for fleet configuration.

8.8 Sequence Number

The sequence number is a 32-bit counter maintained per Frame Category. Sequence numbers SHALL monotonically increase within the lifetime of a link session. The sequence number is reset only at session refresh.

8.9 Path Identification

A path is identified by the tuple `{ src_nid, dst_nid, group_id, sequence_range }`. The path is active for the duration of the call.

8.10 Route Discovery Frames

Route Discovery uses extended Beacon and Path frames. The path frame carries a path descriptor:

Field	Size	Description
type	1 byte	Path record type.
hop_count	1 byte	Number of hops in path.
metric	4 bytes	Cumulative path metric.
path	32 bytes	Maximum path length of 16 node descriptors.

Each node descriptor is 2 bytes containing the short alias of the node.

9. Medium Access Control

9.1 Goals

The MAC SHALL provide:

- Arbitration for concurrent access to the half-duplex channel.
- Predictable latency for voice traffic.
- Reasonable fairness for data and control traffic.
- Robustness against hidden terminal conditions.

9.2 Frame Classes

The MAC recognizes three frame classes:

- **Voice class.** VOPEN, VFRAME, VACK, VEND. These SHALl have priority over all other traffic.
- **Control class.** Beacon, Probe, Path, CTRL. These SHALL be transmitted in the contention window.
- **Data class.** DATA. These SHALl be transmitted in the data opportunity windows.

9.3 Superframe

The MAC SHALL operate on a 480 ms superframe. The superframe is divided into slots as follows:

Slot	Duration	Type
0	16 ms	Contention window.
1	16 ms	Contention window.
2	8 ms	Data opportunity.
3	8 ms	Data opportunity.
4	8 ms	Data opportunity.
5	8 ms	Data opportunity.
6	8 ms	Data opportunity.
7	8 ms	Data opportunity.
8	8 ms	Data opportunity.
9	8 ms	Data opportunity.
10	8 ms	Data opportunity.
11	8 ms	Data opportunity.
12	8 ms	Data opportunity.
13	8 ms	Data opportunity.
14	8 ms	Data opportunity.
15	8 ms	Data opportunity.
16	8 ms	Data opportunity.
17	8 ms	Data opportunity.
18	8 ms	Data opportunity.
19	8 ms	Data opportunity.
20	8 ms	Data opportunity.
21	8 ms	Data opportunity.
22	8 ms	Data opportunity.
23	8 ms	Data opportunity.
24	8 ms	Data opportunity.
25	8 ms	Data opportunity.

Slot	Duration	Type
26	8 ms	Data opportunity.
27	8 ms	Data opportunity.
28	8 ms	Data opportunity.
29	8 ms	Data opportunity.

The slot boundary timing is referenced from the local clock of the node. The clock is loosely synchronized to neighbors via the beacon procedure.

9.4 CSMA Backoff for Control Class

A node wishing to transmit a control class frame SHALL:

1. Perform CCA for 8 symbol periods.
2. If the channel is busy, wait a random number of slot periods uniformly distributed between 1 and 16.
3. Repeat until the channel is idle or the maximum retry count is exhausted.

The maximum retry count for control frames SHALL be 8.

9.5 Voice Floor Acquisition

When a node initiates a voice spur, it SHALL transmit a VOPEN frame in the contention window. The MAC SHALL treat the VOPEN frame as a request to acquire the voice floor.

Subsequent VFRAME frames SHALL be transmitted in the data opportunity windows. The MAC SHALL transmit VFRAME frames in the earliest opportunity after the source has the epoch ready.

A node SHALL yield the voice floor upon transmitting a VEND frame.

9.6 Voice Floor Preemption

A node with higher priority SHALL be permitted to preempt a lower-priority voice spur. Priority is determined by group registration order and explicit priority bits. Preemption is signaled by an unsolicited VOPEN frame and is recorded by the lower-priority talker as a spur termination.

9.7 Relay Arbitration

A node that is forwarding voice traffic on behalf of another node SHALL respond to the prediction of the next hop arrival. The relay node SHALL have higher priority for the next opportunity than the talker for the data opportunity windows immediately following the predicted arrival.

9.8 Data Service Window

Data frames are transmitted in the data opportunity windows. The MAC SHALL follow the same CSMA procedure as control frames, with a maximum retry count of 16.

9.9 Channel Reservation

A node MAY reserve a sequence of data opportunity windows by transmitting a CTRL frame with the `reserve` flag set. The reservation is valid for the next 16 windows. The reservation is renewed only by re-transmission of the CTRL frame.

9.10 Timing Precision

The MAC SHALL maintain a timestamp with resolution of 1 µs. Slot boundaries SHALL be measured relative to the local timestamp. Drift of up to 50 ppm is tolerated.

10. Routing and Path Maintenance

10.1 Goals

The routing layer SHALL provide:

- Discovery of mesh nodes without prior configuration.
- Maintenance of routing tables for up to 32 nodes with sub-50 ms failover.
- Multipath forwarding with primary and backup next hops.
- Loop-free forwarding.

10.2 Routing Table

Each node maintains a routing table with the following entries:

Field	Size	Description
dst_nid	4 bytes	Destination NID.
primary_next_hop	2 bytes	M16 of primary next hop.
backup_next_hop_1	2 bytes	M16 of backup next hop.
backup_next_hop_2	2 bytes	M16 of backup next hop.
primary_metric	4 bytes	Cumulative metric to primary.
backup_metric	4 bytes	Cumulative metric to backup.
last_heard	4 bytes	Timestamp of last successful receive.
age	2 bytes	Age of entry.
flags	1 byte	Routing flags.

Entries are aged out after 30 seconds of no reception.

10.3 Path Metric

The path metric is a 32-bit unsigned integer. The metric shall be computed as:

```
metric = (link_quality * 1000) + (hop_count * 100) + (mcs_penalty * 10)
```

where:

- `link_quality` is the inverse PER multiplied by 256.
- `hop_count` is the number of hops in the path.
- `mcs_penalty` is an MCS-dependent penalty factor that biases selection toward robustness.

The MCS penalty table is:

MCS	Penalty
0-4 (range)	0
5-7 (mesh)	5
8-9 (burst)	20
10-11 (fast)	50

Lower metric values are preferred.

10.4 Originator Messages

A node SHALL transmit an OGM every 1 second on idle. The OGM contains:

Field	Size	Description
type	1 byte	Set to 0x00.
src_nid	4 bytes	Source NID.
seq	4 bytes	OGM sequence number.
metric	4 bytes	Local metric.
neighbors	8 bytes	Up to 4 neighbor descriptors.

Each neighbor descriptor is 2 bytes containing the M16 of the neighbor.

10.5 Bidirectional Reachability

A neighbor SHALL be considered bidirectionally reachable only when both nodes have observed each other's OGM within the last 2 seconds. A one-way observed neighbor SHALL be flagged as unidirectional but still maintained for monitoring.

10.6 Path Update

On receipt of an OGM, a node SHALL update its routing table as follows:

1. If the OGM is from a newly seen node, insert a new entry.
2. If the OGM is from a known node, update the existing entry.
3. If the OGM provides a path with a better metric, update the entry.
4. If the OGM is from a neighbor, update the bidirectional flag.

10.7 Path Tear-Down

A node SHALL mark a path as torn down if no reception has occurred for 30 seconds. The entry SHALL be retained for 5 additional seconds to allow graceful teardown, then deleted.

10.8 Route Repair

If the primary next hop for a destination becomes unavailable, the node SHALL switch to the first backup next hop. If no backup is available, the node SHALL initiate a route repair by transmitting a Path frame with the route request flag set.

The route request is propagated with a TTL of 4 hops. The first responder becomes the new primary next hop.

10.9 Path Establishment for Voice

When a voice call is initiated, the source node SHALL ensure that a path to the group exists. If no path exists, the source SHALL initiate a route repair for the group. If the repair fails, the call SHALL be terminated with a "no route" indication.

10.10 Shortest Path Selection

The path selection algorithm uses Dijkstra's algorithm with the metric defined in Section 10.3. The algorithm is executed upon topology changes.

10.11 Loop Prevention

Each frame SHALL carry a hop count. The hop count SHALL be incremented on each forward. A frame SHALL be discarded if the hop count exceeds 16.

11. Voice Service

11.1 Goals

The voice service SHALL provide:

- Codec2 voice encoding at 3,200 bps or 1,600 bps.
- Forward error correction for the voice payload.
- Playout with bounded latency.
- Integration with the completion protocol.

11.2 Voice Codec Modes

The voice service supports the following Codec2 modes:

Mode	Bit Rate	Frame Length
V0	3,200 bps	20 ms
V1	1,600 bps	40 ms

V0 is the default voice mode.

V1 is permitted for very weak links or extremely dense meshes.

11.3 Voice Frame Structure

A VFRAME SHALL encapsulate the following:

Field	Size	Description
call_id	8 bytes	Call identifier.
epoch	4 bytes	Epoch sequence number.
codec_mode	1 byte	0x00 = V0, 0x01 = V1.
voice_data	variable	Codec2 encoded voice.
fec_block	variable	Reed-Solomon outer FEC.

The Reed-Solomon parameters are:

- GF(2^8) with primitive polynomial 0x11D.
- nsym = 16 bytes for V0, 32 bytes for V1.

11.4 Voice Spur Lifecycle

A voice spur SHALL proceed through the following states:

1. **Idle.** The voice service is not active.
2. **VOPEN.** The VOPEN frame has been transmitted. Awaiting floor grant.
3. **Active.** VFRAME frames are being transmitted.
4. **VEND.** The VEND frame has been transmitted. Awaiting repair tail.
5. **Repair.** Repair transmissions are in progress.

6. **Closed.** The spur is closed.

11.5 VOPEN Frame

The VOPEN frame SHALL include:

Field	Size	Description
call_id	8 bytes	New call identifier.
group_id	2 bytes	Target group.
codec_mode	1 byte	Codec2 mode.
mcs	1 byte	MCS for the call.
flags	1 byte	Call flags.
spur_id	4 bytes	Spur identifier.

11.6 VFRAME Transmission

The voice service SHALL queue VFRAME frames for transmission as soon as the upstream encoder produces a new epoch. The MAC SHALL schedule the VFRAME for the next available data opportunity window.

11.7 VFRAME Reception

On receipt of a VFRAME, the voice service SHALL:

1. Verify the AEAD tag.
2. Check the call_id and seq_number against the spur record.
3. Insert the epoch into the jitter buffer.
4. Update the completion protocol.

11.8 Jitter Buffer

The voice service SHALL maintain a jitter buffer of 3 to 5 epochs. The depth SHALL be dynamically adjusted based on observed jitter.

11.9 Packet Loss Concealment

If the jitter buffer underflows, the voice service SHALL generate concealed audio using Codec2's built-in PLC.

11.10 VEND Frame

The VEND frame SHALL signal the end of a spur. It SHALL include:

Field	Size	Description
call_id	8 bytes	Call identifier.
last_epoch	4 bytes	Last epoch number.
spur_id	4 bytes	Spur identifier.

11.11 End-of-Spur Repair

For a configurable period after VEND (default 300 ms), the network SHALL continue to propagate repair bursts for missing epochs. The period SHALL be expressed in the VEND frame.

11.12 VACK Frame

The VACK frame SHALL include a bitmap of confirmed epochs. The bitmap is 32 bits in length, indicating confirmation status for the last 32 epochs.

11.13 VACK Aggregation

A VACK frame MAY be aggregated with other VACK frames from different sources. The aggregated VACK frame contains a list of {src_nid, bitmap} pairs.

12. Data Service

12.1 Goals

The data service SHALL provide:

- Reliable datagram delivery with optional streaming semantics.
- Fragmentation and reassembly for datagrams up to 8,192 bytes.
- Endpoint to endpoint acknowledgments.
- Best-effort or reliable delivery modes.

12.2 Data Frame

The DATA frame SHALL consist of:

Field	Size	Description
src_nid	4 bytes	Source NID.
dst_nid	4 bytes	Destination NID.
seq	4 bytes	Sequence number.
frag	1 byte	Fragment information.

Field	Size	Description
flags	1 byte	Data flags.
length	2 bytes	Payload length.
payload	variable	Payload data.

The `frag` field indicates whether the frame is the first, middle, last, or solo fragment.

12.3 Data Service Operation

A node initiating a data transfer SHALL:

1. Open a data session with the destination.
2. Fragment the payload into DATA frames.
3. Transmit the frames in the data opportunity windows.
4. Wait for acknowledgments.
5. Retransmit unacknowledged frames up to 7 times.

12.4 Data Acknowledgement

An ACK MAY be transmitted in response to a DATA frame. The ACK indicates the sequence number ranges that have been received.

12.5 Data Reliable Mode

In reliable mode, the source SHALL retransmit unacknowledged frames. The retransmission SHALL be scheduled with exponential backoff.

12.6 Data Best-Effort Mode

In best-effort mode, the source SHALL transmit each frame once. No retransmission is performed.

12.7 Data Service Limits

The maximum datagram size SHALL be 8,192 bytes. The maximum fragment size SHALL be 200 bytes. The maximum outstanding data frames SHALL be 32.

12.8 Data Service Priority

Data frames SHALL have lower priority than voice frames. Data frames MAY be preempted by voice traffic.

13. Group Completion Protocol

13.1 Goals

The Group Completion Protocol (GCP) SHALL provide:

- Confirmation of voice delivery to group members.
- Repair of lost epochs.
- Explicit fail indication to the talker.

13.2 Group Membership

A node SHALL maintain a group membership table. The table SHALL be updated through the discovery protocol. The table SHALL contain:

Field	Size	Description
gid	2 bytes	Group identifier.
member_nid	4 bytes	Member NID.
member_alias	2 bytes	Member M16.
last_seen	4 bytes	Timestamp of last contact.
rssi	1 byte	Last RSSI.
flags	1 byte	Membership flags.

Members are removed from the table after 60 seconds of no contact.

13.3 Confirmation Bitmap

Each node SHALL maintain a confirmation bitmap per spur. The bitmap is 32 bits in length, indicating confirmation status for the last 32 epochs.

The bitmap SHALL be updated as frames are received.

13.4 HAVE Vector Exchange

A node SHALL periodically broadcast its HAVE vector for active spurs. The broadcast SHALL be aggregated with the beacon at a frequency of 1 Hz.

13.5 Aggregation

A receiving node SHALL aggregate the HAVE vectors from multiple neighbors into a per-spur bitmap. The bitmap is forwarded upstream toward the talker along the reverse path.

13.6 Repair Initiation

The talker SHALL initiate a repair for an epoch if:

1. The epoch has not been confirmed by at least 80% of the enrolled group.
2. The expected confirmation deadline has been reached.

The expected confirmation deadline SHALL be 200 ms after the epoch was transmitted.

13.7 Repair Transmission

A repair MAY be implemented as:

- **Direct retransmission.** The original epoch is retransmitted.
- **Fountain code.** A random linear combination of the unreceived epochs is transmitted.

Fountain coding is RECOMMENDED for spurs with multiple lost epochs.

13.8 Repair Algorithm

The repair algorithm SHALL be:

```
while missing_epochs > 0 and retry_count < max_retries:
    if missing_epochs == 1:
        transmit missing epoch directly
    else:
        generate random linear combination
        transmit combination
    retry_count += 1
```

The maximum retry count SHALL be 16.

13.9 Completion Status

The talker SHALL compute the completion status at the end of the repair tail. The completion status SHALL be:

- **Complete.** All enrolled members have confirmed receipt of the spur.
- **Partial.** Some enrolled members have confirmed receipt.
- **Failed.** No enrolled members have confirmed receipt, or no repair was possible.

The completion status SHALL be displayed on the talker's UI.

13.10 Completion Notification

The talker SHALL emit a notification upon completion status change. The notification SHALL include:

- The completion status.
- The list of confirmed members.

- The list of missing members.

14. Operations, Administration, and Maintenance

14.1 Goals

The OAM layer SHALL provide:

- Configuration of mesh parameters.
- Monitoring of mesh health.
- Diagnostics and troubleshooting.
- Firmware update.

14.2 Configuration

A node SHALL expose the following configuration parameters:

Parameter	Description
NID	The node identifier.
M16	The short alias.
GID list	The list of subscribed groups.
Master key	The device master key.
Group keys	The set of group pre-shared keys.

Configuration MAY be performed via the local UI, via OAM frames, or via a wired interface.

14.3 Monitoring

A node SHALL expose the following monitoring metrics:

- RSSI per neighbor.
- PER per neighbor.
- Path metric to each known node.
- Frame error rate.
- Battery state.
- Temperature.

14.4 Diagnostics

A node SHALL provide the following diagnostics:

- Frame log.
- Path trace.
- RSSI history.
- Latency histogram.

14.5 Firmware Update

A node SHALL support firmware update via the data service. Firmware updates SHALL be authenticated by the device master key.

14.6 Reset

A node SHALL provide a reset procedure that restores the factory configuration.

14.7 Time Source

The node MAY be synchronized to an external time source. The protocol is independent of the time source.

14.8 Logging

A node SHALL log significant events. The log MAY be accessed via the local UI or via OAM frames.

15. Compliance Profiles

15.1 Profile Classes

Mesh15 defines the following profile classes:

- **M0.** Plainvoice plain. No cryptography. Minimum compliance.
- **M1.** Minimum compliance. Cryptography. Discovery.
- **M2.** Full compliance. All features.

15.2 Ham Profile

The Ham profile is a compliance profile for amateur radio use. The Ham profile SHALL:

- Disable payload AEAD.
- Emit a periodic beacon containing the operator's callsign.
- Restrict transmit to amateur radio bands.
- Limit occupied bandwidth to the configured channel.

15.3 Business and Public Safety Profile

The Business and Public Safety profile is a compliance profile for commercial and public safety use. The Business profile SHALL:

- Enable payload AEAD.
- Emit a periodic beacon containing the unit identifier.
- Restrict transmit to authorized frequencies.
- Enforce cryptographic group membership.

15.4 SAR Profile

The SAR profile is a compliance profile for search and rescue use. The SAR profile is a superset of the Business profile.

15.5 Profile Selection

A node SHALL select its profile at startup. The profile SHALL be persisted across resets.

16. Performance Budgets and Engineering Limits

16.1 Link Budget

The reference link budget for the HT-15 platform is:

Parameter	Value
TX power	+33 dBm at antenna connector
RX antenna effective	-4 dBi
Body loss	4 dB
RF chain loss	1.5 dB
Fade margin (90% reliability)	10 dB
2 m handheld range MCS-M0	approximately 2 km
2 m handheld range MCS-M1	approximately 1.1 km
70 cm handheld range MCS-M0	approximately 1.2 km
70 cm handheld range MCS-M1	approximately 0.7 km

The exact values are environment-dependent and SHALL be verified by field measurement.

16.2 End-to-End Latency

The reference end-to-end latency for a voice epoch is:

Hop Count	MCS	Latency
1	MCS-M1	80 ms
2	MCS-M1	130 ms
3	MCS-M1	180 ms
4	MCS-M1	240 ms
5	MCS-M1	300 ms

These values include one half-duplex turn per hop, FEC decoding, and a 3-epoch jitter buffer.

16.3 Mesh Capacity

The maximum recommended mesh size is 32 nodes for full voice QoS. Larger meshes SHALL continue to operate with degraded service.

16.4 Maximum Frame Length

The maximum frame length SHALL be 255 bytes inclusive of all headers and trailers.

16.5 Maximum Payload Length

The maximum payload length SHALL be 251 bytes.

16.6 Maximum Hop Count

The maximum hop count SHALL be 16.

16.7 Maximum Outstanding Frames

The maximum outstanding voice frames per spur SHALL be 32.

16.8 Maximum Spur Duration

The maximum spur duration SHALL be 5 minutes. Spurs exceeding this limit SHALL be terminated with a VEND frame and a new spur initiated.

16.9 Airtime Utilization

Voice traffic for a typical spur SHALL occupy no more than 30% of the available airtime. Data traffic SHALL be throttled to maintain this limit.

16.10 Battery Life

The reference battery life for a 2,200 mAh battery is 8 hours of 5-5-90 duty cycle.

17. Security Considerations

17.1 Threat Model

Mesh15 is designed to operate in adversarial environments. The threat model includes:

- **Passive eavesdropping.** An adversary attempts to capture traffic.
- **Active injection.** An adversary attempts to inject forged frames.
- **Replay.** An adversary attempts to replay captured frames.
- **Routing manipulation.** An adversary attempts to manipulate the routing tables.
- **Denial of service.** An adversary attempts to prevent voice traffic.

17.2 Cryptographic Protections

The cryptographic layer provides:

- Confidentiality via ChaCha20.
- Integrity via Poly1305.
- Replay protection via sequence numbers.
- Identity via Ed25519 signatures.

17.3 Key Management

Key management SHALL be performed via a secure provisioning channel. The provisioning channel is out of scope for this specification.

17.4 Privacy

The cryptographic layer SHALL encrypt the source and destination identifiers in the inner header. The outer header contains only the short alias.

17.5 Replay Protection

The sequence number and the nonce construction SHALL prevent replay attacks.

17.6 Routing Security

Routing security is OPTIONAL. A node MAY sign OGMs with its identity key.

17.7 Denial of Service

The protocol is designed to tolerate loss of nodes. The superframe structure provides for graceful degradation.

17.8 Compliance with Regulatory Restrictions

The protocol SHALL be profiled for the deployment domain. Some deployments SHALL restrict the use of cryptographic features.

17.9 Hardware Tamper Resistance

The protocol assumes that the device identity key is stored in tamper-resistant storage. The protocol is not secure if the storage is compromised.

18. Conformance

18.1 Conformance Levels

A Mesh15 implementation SHALL declare its conformance level. The conformance levels are:

- **M0.** Plainvoice, no cryptography.
- **M1.** Cryptography, minimum.
- **M2.** Full.

18.2 M0 Requirements

An M0 implementation SHALL:

- Implement the PHY layer for MCS-R0, MCS-R3, MCS-M0, and MCS-M1.
- Implement the MAC layer.
- Implement the routing layer.
- Implement the voice service in V0 mode.
- Implement the group completion protocol.

18.3 M1 Requirements

An M1 implementation SHALL satisfy all M0 requirements and SHALL:

- Implement the cryptographic layer.
- Implement the data service.

18.4 M2 Requirements

An M2 implementation SHALL satisfy all M1 requirements and SHALL:

- Implement the V1 voice mode.
- Implement the burst profile.
- Implement the fast profile.
- Implement the fountain repair.

18.5 Conformance Test Vectors

A conforming implementation SHALL pass the conformance test vectors published in Appendix C.

18.6 Conformance Identifier

A conforming implementation SHALL expose a version string of the form `MESH15-x.y.z-c` where x.y.z is the protocol version and c is the conformance level.

19. Revision History

Version	Date	Description
0.1.0	Initial	Initial draft.

Appendix A: PHY Register Tables

The following tables define the CC1200 register settings for each MCS. The values are normative for interoperability.

(Detailed register tables are provided in Annex A of this specification as a separate document.)

Appendix B: Frame Examples

The following examples illustrate the frame construction.

B.1 VFRAME Example

A VFRAME with MCS-M1, group 0x0001, call_id 0x123456789ABCDEF0, epoch 0x00000001, V0 codec:

```
Outer Header:
  version     = 0x01
```

```
    frame_type  = 0x04
    flags       = 0x00
    mcs         = 0x06
    src_alias   = 0x1234
    dst_alias   = 0xFFFF

  Inner Header:
    src_nid     = 0xABCD1234
    dst_nid     = 0xFFFFFFFF
    group_id    = 0x0001
    seq_number  = 0x00000001
    hop_count   = 0x00
    flags       = 0x00

  Payload:
    call_id     = 0x123456789ABCDEF0
    epoch       = 0x00000001
    codec_mode  = 0x00
    voice_data  = 8 bytes (V0 epoch)
    fec_block   = 16 bytes (Reed-Solomon)

  MIC:
    16 bytes (Poly1305)
```

B.2 VOPEN Example

A VOPEN with MCS-M0, group 0x0001, V0 codec:

```
  Outer Header:
    version     = 0x01
    frame_type  = 0x03
    flags       = 0x00
    mcs         = 0x05
    src_alias   = 0x1234
    dst_alias   = 0xFFFF

  Inner Header:
    src_nid     = 0xABCD1234
    dst_nid     = 0xFFFFFFFF
    group_id    = 0x0001
    seq_number  = 0x00000000
    hop_count   = 0x00
    flags       = 0x00

  Payload:
    call_id     = 0x123456789ABCDEF0
    group_id    = 0x0001
    codec_mode  = 0x00
    mcs         = 0x05
    flags       = 0x00
    spur_id     = 0x00000001

  MIC:
    16 bytes
```

Appendix C: Conformance Test Vectors

(Documented in a separate file.)

Appendix D: Glossary

Term	Definition
AEAD	Authenticated Encryption with Associated Data.
Beacon	A periodic frame used for discovery.
BMP	Bitmap.
Bore	See Spur.
Elder	A node with high path rank.
Epoch	A single block of voice data.
Frame	The atomic unit of transmission.
Fountain	A random linear code, also known as RaptorQ.
OGM	Originator Message.
Playout	The act of rendering an audio stream.
Spur	A continuous talkspurt.
Stub	A node with only one path.

Appendix E: Bibliographic References

Reference	Description
RFC 2119	Key words for use in RFCs to Indicate Requirement Levels.
RFC 2104	HMAC: Keyed-Hashing for Message Authentication.
RFC 5869	HMAC-based Extract-and-Expand Key Derivation Function.
RFC 7539	ChaCha20 and Poly1305 for IETF Protocols.
RFC 7748	Elliptic Curves for Security.
IETF Ed25519	Edwards-curve Digital Signature Algorithm.
TI CC1200	Texas Instruments CC1200 Low-Power High-Performance Sub-1 GHz RF T
Codec 2	David Rowe, Codec 2.

Appendix F: Acknowledgments

This specification is the product of extensive design effort. The contributors to the Mesh15 protocol include the engineering team at Arkos Engineering and the open-source community.

Appendix G: Document Index

The index is provided as a separate file.

Appendix H: M17 Interoperation Profile (Reserved)

A future annex will define an optional M17 interoperation profile.

Appendix I: Beacon and OGM Timeslot Allocation

The beacon and OGM transmission timeslot is allocated as follows:

- Beacon: 1 second periods.
- OGM: 1 second periods.

Each beacon SHALL be sent in the contention window.

Appendix J: Coding Procedures

The coding procedures are normative. The implementation SHALL follow the procedures described in Section 6.

Appendix K: Compliance Test Procedures

The compliance test procedures are normative. The implementation SHALL pass the tests described in this section.

Appendix L: Path Selection Worked Example

This appendix provides a worked example of the path selection algorithm.

Appendix M: Repair Worked Example

This appendix provides a worked example of the repair algorithm.

Appendix N: Frame Bitmap Specifications

This appendix provides the formal specification of the various bitmaps used in the protocol.

Appendix O: Voice Playout Worked Example

This appendix provides a worked example of the voice playout algorithm.

Appendix P: Discovery Protocol Worked Example

This appendix provides a worked example of the discovery protocol.

Appendix Q: Encryption Procedure Worked Example

This appendix provides a worked example of the encryption procedure.

Appendix R: Routing Table Worked Example

This appendix provides a worked example of the routing table construction.

Appendix S: Membership Update Procedure

This appendix provides the formal specification of the membership update procedure.

Appendix T: Time Synchronization Procedure

This appendix provides the formal specification of the time synchronization procedure.

Appendix U: Clock Drift Compensation

This appendix provides the formal specification of the clock drift compensation procedure.

Appendix V: Power Conservation Procedure

This appendix provides the formal specification of the power conservation procedure.

Appendix W: Antenna Diversity Procedure

This appendix provides the formal specification of the antenna diversity procedure.

Appendix X: Frequency Hopping Procedure

This appendix provides the formal specification of the frequency hopping procedure.

Appendix Y: Simplex/Duplex Mode Procedure

This appendix provides the formal specification of the simplex and duplex mode procedures.

Appendix Z: Compliance Tests

The compliance tests are described in Annex C of this specification.

Annex A: PHY Register Tables

This annex provides the detailed CC1200 register tables for each MCS. The tables are normative for interoperability.

A.1 MCS-R0: 2-GFSK 1.2 kbps

Register	Value	Description
SYMBOL_RATE2	0x5F	1.2 ksym/s
SYMBOL_RATE1	0x75	mantissa
SYMBOL_RATE0	0x10	mantissa

Register	Value	Description
MODCFG_DEV_E	0x01	2-GFSK, dev_e=1
DEVIATION_M	0x06	5 kHz deviation
CHAN_BW	0x02	12.5 kHz
PKT_CFG0	0x05	variable length
PKT_LEN	0xFF	max length
PA_CFG1	0x3F	+33 dBm
PA_CFG0	0x42	ramp

(Additional register values are provided in this annex.)

A.2 MCS-R1: 2-GFSK 2.4 kbps

(Register table provided in this annex.)

A.3 MCS-R2: 2-GFSK 4.8 kbps

(Register table provided in this annex.)

A.4 MCS-R3: 2-GFSK 9.6 kbps

(Register table provided in this annex.)

A.5 MCS-R4: 4-GFSK 9.6 kbps

(Register table provided in this annex.)

A.6 MCS-M0: 2-GFSK 19.2 kbps

(Register table provided in this annex.)

A.7 MCS-M1: 2-GFSK 38.4 kbps

(Register table provided in this annex.)

A.8 MCS-M2: 2-GFSK 50 kbps

(Register table provided in this annex.)

A.9 MCS-B0: 2-GFSK 100 kbps

(Register table provided in this annex.)

A.10 MCS-B1: 2-GFSK 250 kbps

(Register table provided in this annex.)

A.11 MCS-F0: 2-GFSK 500 kbps

(Register table provided in this annex.)

A.12 MCS-F1: 2-GFSK 1 Mbps

(Register table provided in this annex.)

Annex B: Conformance Test Vectors

The conformance test vectors are provided in this annex. Each test vector includes:

- Input parameters.
- Expected output.
- Edge cases.

(This annex is published as a separate document.)

Annex C: Reference Implementation Notes

The reference implementation notes are provided in this annex. The notes describe the implementation choices made in the reference implementation.

Index of Tables

Index of Figures

End of Specification

This document is the canonical specification of the Mesh15 protocol.